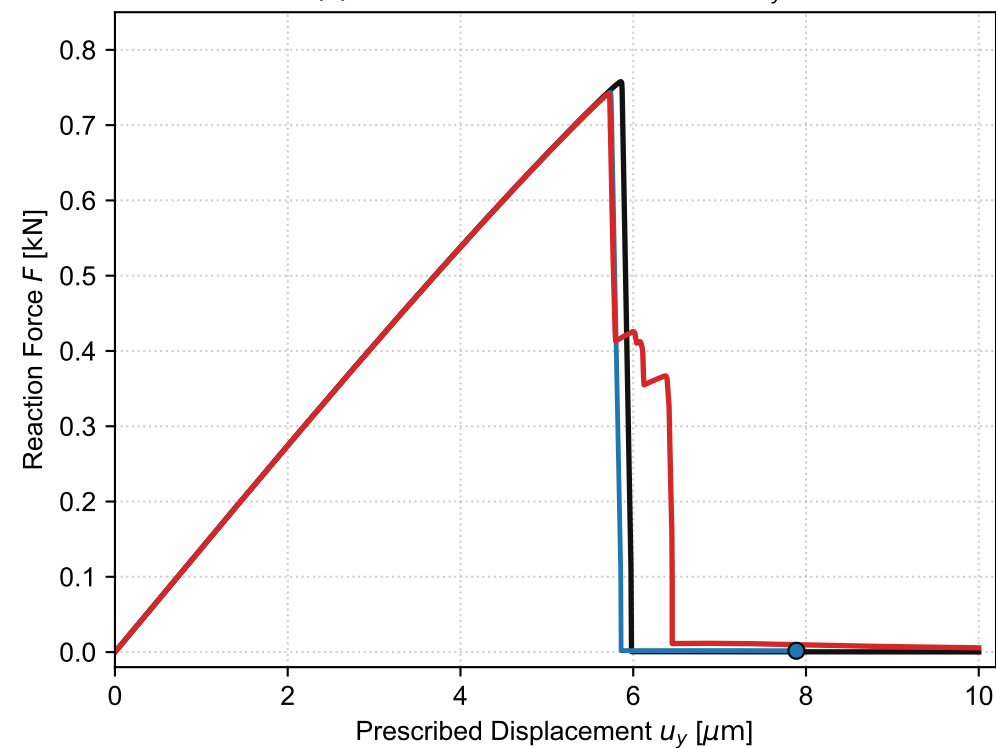
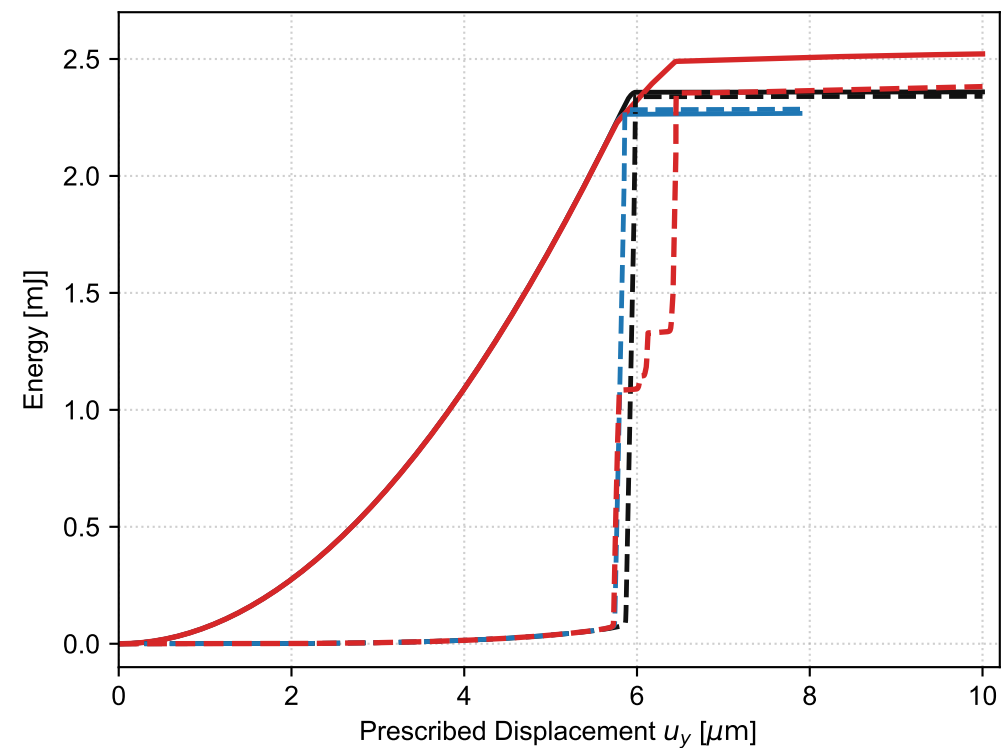
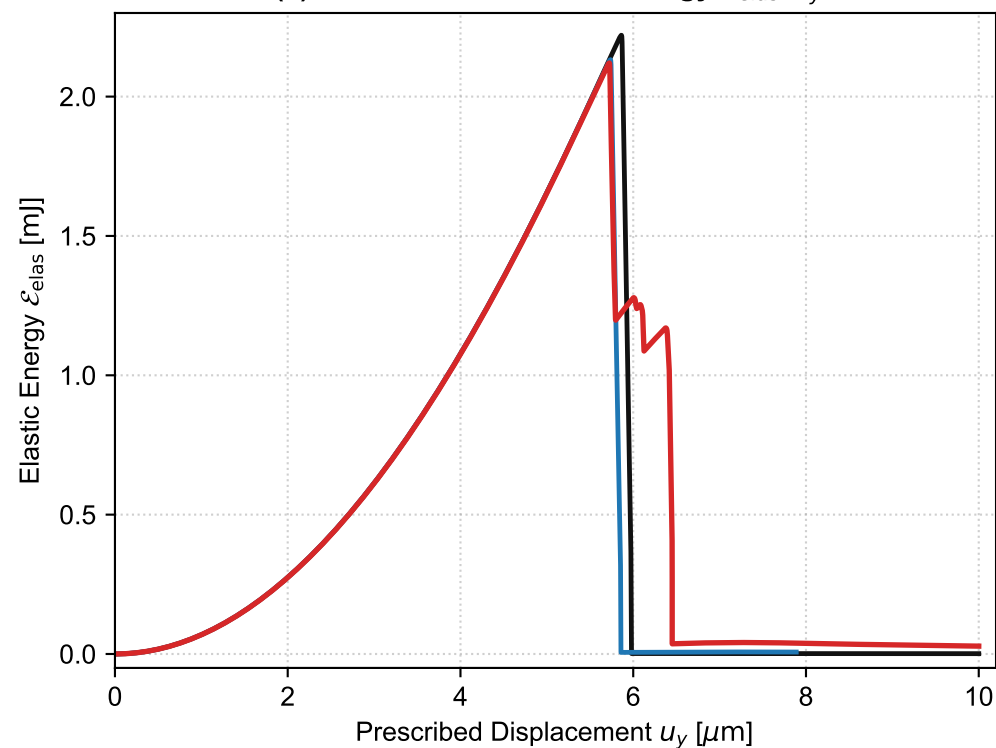


(a) Structural Reaction Force $F(u_y)$ 

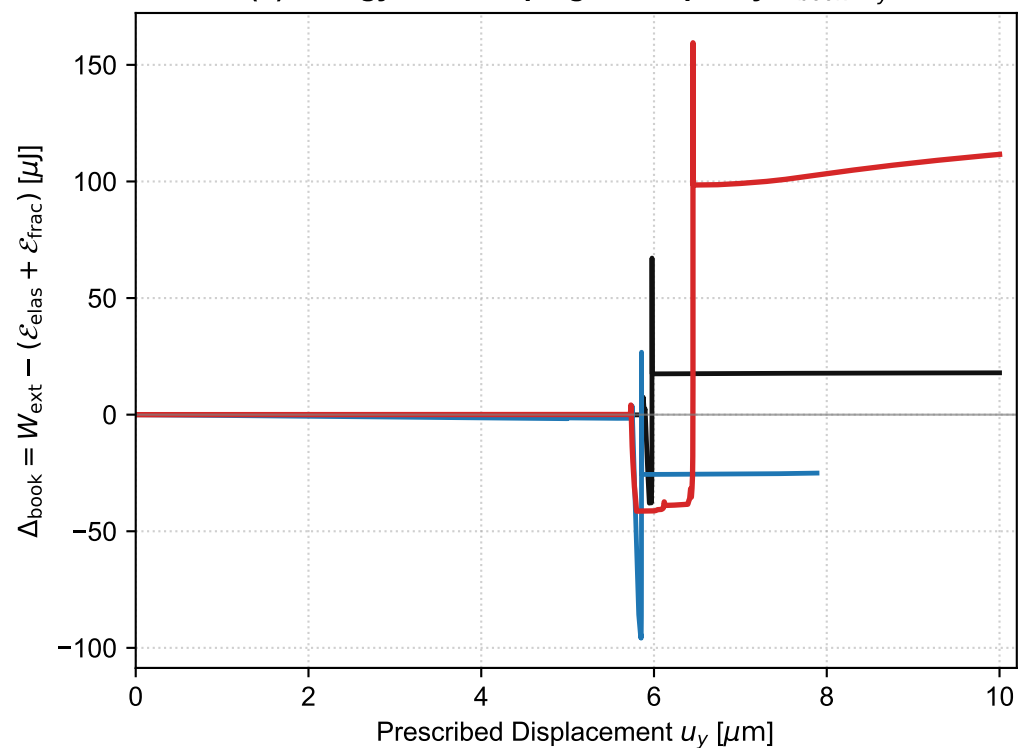
- Fixed benchmark (15,192 FE)
- Corrected ET1 (14,483 FE)
- Spatial-fine adaptive (57,929 FE)

(b) External Work W_{ext} & Fracture Surface Functional $\mathcal{E}_{\text{frac}}$ 

- W_{ext} Benchmark 15k
- W_{ext} Adaptive ET1 14k
- W_{ext} Fine Adapt 58k
- $\mathcal{E}_{\text{frac}}$ Benchmark 15k
- $\mathcal{E}_{\text{frac}}$ Adaptive ET1 14k
- $\mathcal{E}_{\text{frac}}$ Fine Adapt 58k

(c) Stored Elastic Strain Energy $\mathcal{E}_{\text{elas}}(u_y)$ 

- Fixed Benchmark Anchor 15k
- Corrected Adaptive ET1 14k
- Spatial-Fine Adaptive Ref 58k

(d) Energy Bookkeeping Discrepancy $\Delta_{\text{book}}(u_y)$ 

- Fixed Benchmark 15k (Δ_{book} , 0.76%)
- Adaptive ET1 14k (Δ_{book} , 1.10%)
- Spatial-Fine Adaptive 58k (Δ_{book} , 4.43%)